Memory Match: Patch Notes

Version 2 (`index_v2.html`), 2026-10-06

Version 2 responds to the class feedback survey on the first prototype (`index.html`, which is kept unchanged as version 1). There were **29 responses**. Each suggestion was:

1. grouped with similar suggestions,
2. checked against the Hard Constraints in `CLAUDE.md`, and
3. reviewed by the instructor, who chose which changes to build.

Summary: 10 changes were made. 4 workable suggestions were not chosen for this version. 2 ways of reading the feedback could not be built because they break our rules, and 3 comments could not be acted on.

✅ Changes made

1. The whole board fits on one screen (no scrolling)

- **Feedback:** 5 students had to scroll to see all the cards, especially in Hard mode.
- **Change:** The page is now exactly one screen tall. The cards resize to fill the space left under the controls, so every row and column is always visible.
- **Tested:** All three levels fit with no scrolling on a 1280×720 projector screen, a laptop, and a 375-px-wide phone. On extremely short windows the board stops shrinking at a minimum size and the page scrolls instead, so cards never become too small to use.

2. Bigger pictures

- **Feedback:** 1 student asked for bigger images.
- **Change:** The picture now fills about half of each card's width and grows with the card. On large screens cards can be up to 180 px wide (version 1 capped them at about 130 px).

3. Faster flip-back after a wrong pair

- **Feedback:** 4 students found the wait after a wrong pair too long. One asked for a click to flip the cards back instantly.
- **Change:**

    - The wait is shorter: **0.7 seconds**, down from 1 second.
    - **Clicking any card during the wait** flips the wrong pair back immediately and turns over the card you clicked, so fast players never have to wait.
- **Note:** one student thought the delay lasted several seconds. It was actually 1 second, so the wait *felt* longer than it was. That's useful to know when designing game timing.

4. Flip animation

- **Feedback:** 4 students asked for animations, transitions, or effects when flipping.
- **Change:** Cards now turn over with a 3D flip, made with CSS only (no images or libraries).
- **Accessibility:** if a device has "reduce motion" turned on, cards flip instantly instead. Some people get dizzy or sick from on-screen motion.

5. Sound effects, with a mute button

- **Feedback:** 2 students asked for sound effects.
- **Change:** Short beeps play for a flip, a match, a miss, and a win. A 🔊 **Sound on** / 🔇 **Sound off** button turns them off, which matters in a quiet classroom. The game remembers the choice.
- **How it fits our rules:** the sounds are generated in code by the browser's built-in audio feature, so there are **no sound files** to download or credit.

6. Best-scores table (saved on this device only)

- **Feedback:** 3 students asked for a high-score or score table.
- **Change:** The win screen shows your **top 5 scores for that difficulty**: moves, time, points, and theme. A new score is highlighted, shown in bold, and labeled "new!".
- **Ranking:** fewest moves first. Time only breaks ties, because moves are the goal (see #7).
- **How it fits our rules:**
    - Scores are saved **only in this browser on this device**. Nothing is sent anywhere.
    - **No names are asked for or stored.** The table is simply "Your best scores".
    - If the browser blocks saving (e.g. private browsing), the game still works. It shows a note that scores will reset when the page is closed.

7. Moves are the main goal, with a star rating

- **Feedback:** 1 student asked to make moves (not time) the main objective, and 1 asked for a reward for playing.
- **Change:**

- **Moves** is now the largest number on screen. Time is shown smaller.
- The win screen awards **1 to 3 stars** based on moves per pair:
  - ★★★ for 1.75 moves per pair or fewer
  - ★★ for 2.5 or fewer
  - ★ otherwise

  Even perfect memory needs about 1.6 moves per pair on average, because the first flips are blind guesses, so three stars is hard but possible.
- The intro line now says: "Fewer moves earn more stars!"
- The stars are also written out in words ("2 of 3 stars") for screen readers.

8. Points with a streak bonus

- **Feedback:** 2 students suggested a point system, including points for matches in a row.
- **Change:** Each match earns **10 points × your current streak**: 10 for the first match in a row, 20 for the second, 30 for the third, and so on. A wrong pair resets the streak. A message under the stats announces each result, e.g. "Match! +30 points (3 in a row 🔥)".

9. Pictures that are easier to tell apart

- **Feedback:** 3 students found some animals too similar (e.g. the fox, tiger, and lion faces).
- **Change:** The animal set was replaced. Version 1 used 12 similar-looking *faces*. Version 2 uses 12 animals with very different shapes and colors: 🐘 🦒 🦕 🐙 🦋 🐝 🦀 🐌 🦉 🐠 🐋 🦩.

10. Name labels on cards

- **Feedback:** 1 student suggested a number in the corner of each card, to tell look-alike pictures apart.
- **Change:** Each face-up card shows its **name** under the picture (e.g. "Owl"). A name says more than a number would, and screen readers now announce "Owl" instead of reading out the emoji.
- **Small screens:** labels hide themselves on very small cards (Hard mode on a phone), where they would be too small to read.

11. Theme picker: Animals, Fruit, or Numbers

- **Feedback:** 5 students wanted customizable card pictures, and 1 suggested numbers instead of pictures to make the game harder.
- **Change:** A **Theme** menu offers three built-in sets: **Animals**, **Fruit**, and **Numbers** (1–12). Changing the theme deals a new game. The best-scores table records which theme each score was played with.
- **How it fits our rules:** every set is built into the file, so nothing is uploaded or downloaded. (See "Not possible under our rules" for why uploading your own pictures isn't offered.)

> Changes 1 and 2 (fit to screen + bigger pictures) were chosen as one item, so these 11 sections cover the 10 changes the instructor selected.

⏸️ Workable, but not chosen for version 2

These suggestions fit our rules, but the instructor did not pick them for this version. They remain candidates for a future version.

Suggestion	Students	Notes for a future version
Two-player mode (take turns; a match earns another turn)	2	Would be local only: "Player 1" and "Player 2" on one screen. Medium effort.
Type your own card pairs	part of the 5 asking for custom content	Text would stay in the browser. Students would need a reminder not to type personal details. Larger effort.
"How to play" instructions before the first game	1	Small effort. The one-line intro and the new on-screen messages partly cover this.
Custom-drawn artwork instead of emoji	4	Hand-drawn pictures built into the file. Large effort, and would add a lot to the file's length. The more distinct animals and the theme picker address part of this request.

❌ Not possible under our rules

Some feedback could be read in a way that breaks a Hard Constraint in `CLAUDE.md`. In each case we built the version that fits the rules instead.

Suggestion (as it could be read)	Why not	What we did instead
Upload your own images or photos for the cards	**Rule 3 (privacy):** students could upload photos of real people, and photos often carry hidden personal data such as GPS location.	Built-in **theme picker** (#11)
Online multiplayer, or a class-wide leaderboard	**Rule 2 (no internet)** and **Rule 3 (privacy):** this needs a server and players' names.	Private **best-scores table** on each device (#6). A local two-player mode remains a future option.

💬 Feedback that could not be acted on

Comment	Reason
A comment about choosing indicators for a stock analysis	About a different assignment, not this game.
A two-word comment mentioning "issues" with no details	Too vague to identify a problem.
A request to make the game fit when played through the course website	Not a code change: uploading the file to the course site is a sharing decision. Change #1 (fit to screen) should help the game display well inside a course-site window.

How version 2 was checked

- **Gameplay:**
  - Matches, mismatches, and the streak bonus (10 → 20, then back to 10 after a miss).
  - Click-to-skip during a wrong pair.
  - Switching difficulty and theme.

    - The win screen.
- **Layout:** no scrolling on any level at projector, laptop, and phone sizes. Dark mode checked.
- **Saving:**
    - Scores were saved and still there after reopening the file from `file://`.
    - With saving blocked, the game still worked and showed its warning note.
- **Rules scan:** no web addresses, no code that contacts the internet, no outside files. Only non-personal data is saved.
- **Browser console:** no errors.
- **Bug found and fixed during testing:** on phones, the best-scores table was too wide and cut off its last column. It now fits a 375-px screen.

Known limitations

- **Emoji look different on each device** (Apple, Windows, Android, and so on). That's why the name labels help.
- **Best scores belong to one browser on one device.** They don't move between computers, and clearing the browser's data erases them.
- **Sound starts only after the first click.** Browsers block sound until the user interacts with the page.
- **The file is about twice as long as version 1** (about 910 lines vs. 440). It still follows the same labeled sections, but it takes longer to scroll through in a lecture.